

Joint Mitigation of Algorithmic and Physical Errors in Noisy Hamiltonian Simulation

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(Dated:)

Product-formula Hamiltonian simulation is naturally suited to near-term quantum processors, but its accuracy is set by two competing errors: finite-step Trotter bias and physical hardware noise. We introduce a joint extrapolation strategy that ties the tunable noise strength to the Trotter step size, $\lambda(s) = c(sT)^{p+1}$ for a p -th order product formula. Along this one-dimensional path, the leading physical-noise and Trotter terms appear at the same order and can be canceled by a single Richardson extrapolation. The analysis uses a finite-order Baker–Campbell–Hausdorff truncation bound, controlled by doubly right-nested commutators, so that the guarantee does not rely on convergence of the full BCH series. For local Hamiltonians with local Lindbladian noise, the protocol mitigates physical noise together with Trotter error with only a constant asymptotic overhead relative to noiseless Trotter extrapolation, provided the required noise strengths lie above the intrinsic device-noise floor. We demonstrate the workflow on Trotterized Ising dynamics using an adaptively learned sparse Pauli–Lindblad noise strength on a superconducting processor, and benchmark the same one-dimensional construction against a two-dimensional Richardson baseline in tensor-network simulations.

Shuo: To be a physics facing paper

Hamiltonian simulation is a leading route to useful quantum computation, with applications from condensed-matter dynamics to quantum chemistry and high-energy physics [1–4]. Product formulas [5–7] are especially attractive on near-term processors because they compile into repeated local gate layers, require no large ancillary registers, and preserve the locality and commutator structure of the target Hamiltonian. Their experimental difficulty is a depth trade-off: increasing the number of Trotter steps reduces deterministic product-formula bias, but the deeper circuit accumulates more physical noise [8]; using fewer steps protects the device, but leaves a larger algorithmic error. In the regime where near-term experiments are informative, both effects are visible and the relevant target is the joint zero-noise, zero-step limit.

Existing mitigation strategies usually organize this limit as two separate extrapolation directions. Zero-noise extrapolation (ZNE) [9, 10] estimates the zero-noise response at fixed Trotter step number, while Trotter extrapolation reduces the finite-step bias [11, 12]. One may apply these two procedures sequentially, or sample a two-dimensional Richardson grid over noise strength and step size [13]. These approaches are systematic, but they require many circuit families and inherit the conditioning of both extrapolation axes, which is costly precisely when circuit depth is already the limiting resource.

The central idea of this work is to use the depth/noise trade-off as the mitigation coordinate. Instead of sampling the full two-dimensional error surface, we choose a calibrated path through it by tying the tunable noise strength to the Trotter step size, $\lambda(s) = c(sT)^{p+1}$ for a p -th order product formula. Because the leading physical-noise contribution is linear in λ whereas the leading Trotter contribution scales as s^{p+1} , this choice makes both mechanisms appear at the same order in a single variable. A Richardson rule along this one-dimensional path can therefore cancel the leading physical-noise and Trot-

ter terms with one set of coefficients.

The main theoretical question is whether this path retains the commutator-sensitive guarantees that make product formulas useful. Recent work has shown that physical and Trotter errors can be coupled in simplified mitigation settings [14], but a commutator-sensitive analysis for noisy product formulas requires more than a formal power-series argument. We use a finite-order Baker–Campbell–Hausdorff truncation bound controlled by doubly right-nested commutators [15], avoiding any assumption that the full BCH series converges. For local Hamiltonians with local Lindbladian noise, this gives a joint mitigation guarantee with the same asymptotic scaling as noiseless Trotter extrapolation up to constant factors, provided the required noise strengths remain within the experimentally accessible window.

We support this picture in two complementary ways. On superconducting hardware, an adaptively learned sparse Pauli–Lindblad model supplies calibrated noise amplification for the paired extrapolation nodes, demonstrating the one-dimensional workflow on a real device. In tensor-network simulations, the same one-dimensional construction is compared directly with a two-dimensional Richardson baseline, showing comparable extrapolation accuracy while using fewer sampled nodes. Together, these results support a simple physical picture of the protocol: replace a two-axis mitigation problem by a calibrated one-dimensional path, while retaining the commutator structure of the underlying Hamiltonian simulation.

Table I summarizes this positioning relative to existing mitigation routes.

Shuo: Can we jointly mitigate physical and Trotter errors while maintaining commutator scaling?

TABLE I. Qualitative positioning of mitigation strategies for product-formula Hamiltonian simulation.

Feature	ZNE [9, 10]	Trotter extrap. [11, 12]	Sequential/2D [13]	Prior coupled [14]	This work
Mitigates physical noise	✓	×	✓	✓	✓
Mitigates product-formula bias	×	✓	✓	✓	✓
Uses a one-dimensional joint path	×	×	×	✓	✓
Retains provable commutator scaling	×	✓	×	×	✓
Avoids an extra sampling factor	×	×	×	✓	✓

I. RESULTS

XZW addressed: Reader-facing issue: Results should open with the experimental claim and the figure logic, not with the full theory setup. Consider moving most formal definitions below to Methods and keeping only the path geometry plus the takeaway here.

A one-dimensional path for joint extrapolation.

We first present the central idea and theoretical scheme: physical-noise mitigation and Trotter-error extrapolation need not be performed on an independent two-dimensional grid. The scheme calibrates the tunable noise strength λ at each Trotter step size sT and extrapolates along the one-dimensional path Shuo: What is c ?

$$\lambda(s) = c(sT)^{p+1}. \quad (1)$$

This choice matches the leading physical-noise contribution, which is linear in λ , to the leading Trotter contribution, which scales as s^{p+1} . The path is therefore designed so that both mechanisms enter the same Richardson variable. Figure 1 illustrates the geometry for the first-order case $p = 1$, where the coupled path is quadratic in s . The point of the schematic is the Richardson node set: the conventional approach assigns independent nodes to the two axes, while the coupled approach uses each sampled point as one paired node. We then implement the corresponding noise amplification using an adaptively learned sparse Pauli-Lindblad model on a superconducting processor, and finally compare the same one-dimensional path with a two-dimensional Richardson baseline in tensor-network simulations.

The geometric replacement in Figure 1 can be stated as the following informal guarantee; the formal resource theorem, including the Richardson coefficient bounds, finite-BCH conditions, intrinsic-noise-floor assumption and worst-case shot complexity, is given in Supplementary Note 1 [16].

Informal theorem. *If calibrated noise strengths can be realized at the coupled nodes $\lambda(s_j) = c(s_jT)^{p+1}$, then a one-dimensional Richardson rule along this path cancels the leading Trotter and physical-noise biases simultaneously. Thus m paired circuit families replace an $m \times m'$ rectangular Richardson grid, while finite-order BCH expansion preserves the commutator scaling of Trotter error up to constants.*

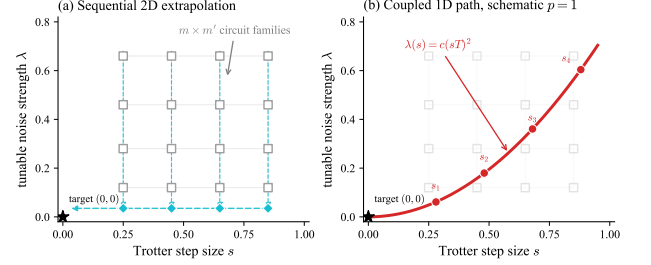


FIG. 1. Coupled extrapolation path as a one-dimensional replacement for a two-dimensional Richardson grid. The horizontal coordinate s is the Trotter step size, the vertical coordinate λ is a tunable physical-noise strength, and the black star marks the ideal target limit $(s, \lambda) = (0, 0)$. (a) A conventional two-axis construction samples an $m \times m'$ rectangular grid, where m and m' denote the Richardson node counts along the step-size and noise-strength axes: one first extrapolates each fixed- s column to zero noise, shown by the cyan diamonds, and then extrapolates those zero-noise estimates toward $s = 0$. (b) The coupled construction instead samples only m paired Richardson nodes on a single path. For the schematic first-order case shown here, the noise is chosen as $\lambda(s) = c(sT)^2$, where T is the total simulated time and c sets the noise scale, so the leading linear-in- λ noise contribution is matched to the leading s^2 Trotter contribution along one Richardson extrapolation variable.

Finite-BCH expansion for error cancellation.

With the noisy product-formula channel $\tilde{\mathcal{P}}(t, \lambda)$ defined in Eq. (9) in Methods, the observable average over the coupled path reduces to a one-variable function

$$f(s) = g(s, \lambda(s)) = \text{Tr} \left[O \tilde{\mathcal{P}}(sT, c(sT)^{p+1})^{1/s} \rho_0 \right]. \quad (2)$$

The function in Eq. (2) is hard to analyze directly because it is defined by a product of noncommuting noisy layers. We therefore introduce a truncated surrogate $f_{q_0}(s)$, obtained by replacing the noisy product channel by its finite-BCH expansion through order q_0 in Eq. (13):

$$f_{q_0}(s) = \text{Tr} \left[O e^{T \mathcal{G}_{(q_0)}(s)} \rho_0 \right] = f(0) + \sum_{\ell=1}^{m-1} a_\ell s^\ell + R_m(s), \quad (3)$$

where $f(0)$ corresponds to the ideal zero-noise, zero-step target, a_ℓ are the coefficients of the low-order polynomial expansion, and $R_m(s)$ is the residual.

For a maximum allowed step size s_0 and all selected nodes $s_j \leq s_0$, the finite-BCH truncation theorem ensures $|f(s_j) - f_{q_0}(s_j)| \leq \varepsilon_{q_0}$, where ε_{q_0} is controlled by doubly right-nested commutators of the Hamiltonian and Lindbladian generators. The deviation of $e^{T\mathcal{G}_{(q_0)}(s)}$ from $e^{-i\text{ad}_H T}$ is expanded by an interaction-picture Dyson series, yielding the power expansion. Then,

$$|f(s) - f(0)| \leq |f(s) - f_{q_0}(s)| + |f_{q_0}(s) - f(0)|.$$

The coupling in Eq. (1) places the leading noise and Trotter contributions at the same order in s , so an m -node Richardson rule cancels the polynomial part $a_\ell s^\ell$ of Eq. (3) with a single set of coefficients, rather than using independent coefficients in a two-dimensional grid.

For Richardson coefficients b_j at nodes s_j , the deterministic error separates as

$$\left| \sum_{j=1}^m b_j f(s_j) - f(0) \right| \leq \|\mathbf{b}\|_1 \varepsilon_{q_0} + \|\mathbf{b}\|_1 \max_{s_j} |R_m(s_j)|. \quad (4)$$

The first term is the accumulated finite-BCH truncation gap between the operational data and the truncated surrogate. The second term is the Richardson error for the surrogate expansion after the low-order polynomial terms cancel. The supplementary material gives the truncation bookkeeping and model-dependent commutator estimates; for k -local Hamiltonians with local Lindbladian noise, these bounds give the same asymptotic scaling as noiseless Trotter extrapolation up to a constant factor in the step size, provided the required noise strengths remain above the intrinsic noise floor.

Experimental demonstration and benchmark.

We demonstrate the mitigation workflow experimentally using transverse-field Ising dynamics,

$$H = -J \sum_{\langle i,j \rangle} Z_i Z_j - h \sum_{i=1}^n X_i, \quad (5)$$

where $\langle i,j \rangle$ denotes the Ising interaction edges in the selected hardware subgraph or in the simulated one-dimensional chain. In the hardware implementation, we equivalently parameterize the transverse-field and interaction strengths by rotation rates $r_x = -h$ and $r_{zz} = -J$. We use a second-order symmetric Trotter formula. The measured observable is the average longitudinal magnetization,

$$O = \frac{1}{n} \sum_{i=1}^n Z_i. \quad (6)$$

The hardware experiments use a connected 9-qubit subgraph of a 12-qubit Wuyue No.2 chip, as shown in Figure 2, while the tensor-network simulations use the same observable on a 20-qubit one-dimensional instance.

For the hardware data, we model the device noise relevant to the compiled circuit family by an effective sparse

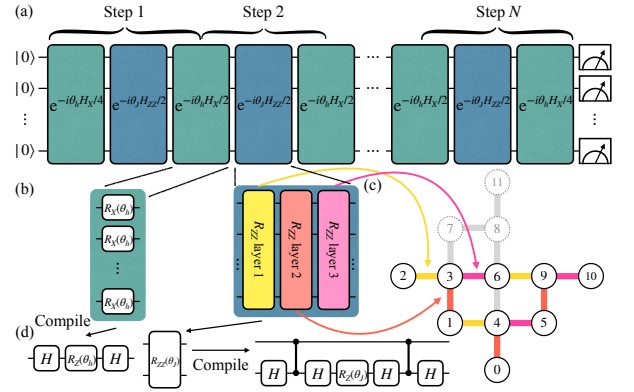


FIG. 2. Hardware implementation of Trotterized Ising dynamics. (a) A second-order symmetric Trotter step alternates transverse-field rotations and Ising interactions. (b) The interaction layer is decomposed into three parallel R_{zz} patterns. (c) A connected 9-qubit subgraph of the 12-qubit device is used; colored edges show the scheduled interaction patterns. (d) Logical R_x and R_{zz} operations are compiled into the native H , R_z , and controlled-Z gates.

Pauli-Lindblad model,

$$\mathcal{L}(\rho) = \sum_{k \in \mathcal{K}} \lambda_k (P_k \rho P_k - \rho), \quad (7)$$

with a retained Pauli set $\mathcal{K}$ much smaller than the full Pauli group. The rates are obtained by an adaptive learning procedure and are used only to set calibrated noise strengths for mitigation, not as a microscopic model of every hardware error process. For each selected Trotter step size s_j , scaling the learned rates and sampling the corresponding Pauli insertions realizes the target noise strength $\lambda(s_j)$ without increasing the depth of the target Trotter circuit. The selected hardware configurations are therefore paired nodes $(s_j, \lambda(s_j))$ of the same one-dimensional extrapolation path used in the theory. Details of the circuit mapping, readout calibration and learned sparse Pauli-Lindblad model are given in Supplementary Notes 3 and 4 [16].

a. Hardware extrapolation. The hardware workflow realizes the coupled path by pairing each Trotter count N_j with the calibrated learned-channel noise strength $\lambda(s_j)$, where $s_j = 1/N_j$. These paired measurements are fitted as a one-variable function of $x = 1/N_{\text{Trotter}}$, exploiting the even-power structure of the second-order product formula. The key experimental check is whether hardware and numerical data move consistently toward the same zero-noise, zero-step reference. This experiment tests the implementability of the coupled one-dimensional extrapolation on hardware in the regime where both physical noise and finite-step error are visible; the tensor-network benchmark below isolates the comparison with a two-dimensional Richardson grid.

Figure 3 shows the expectation values of the average magnetization versus the inverse Trotter steps. Each plotted hardware configuration is a paired node

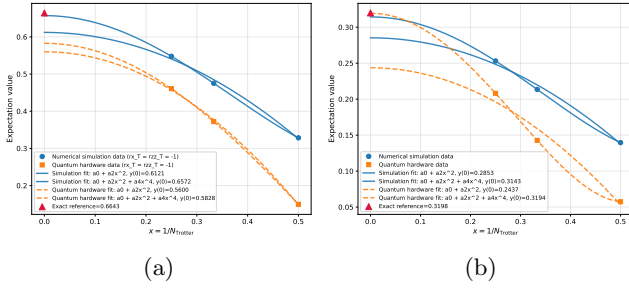


FIG. 3. Joint hardware extrapolation of noise and Trotter errors. Numerical simulation (blue) and superconducting-hardware data (orange) are plotted versus $x = 1/N_{\text{Trotter}}$. The fitted curves extrapolate toward $x = 0$, where both finite-step Trotter error and the amplified physical-noise bias are removed in the fitted model; the red triangle marks the exact reference. Panels show two representative evolution settings: (a) $r_x T = r_{zz} T = -1$ and (b) $r_x T = r_{zz} T = -2$. The compatible hardware and numerical trends support the coupled one-dimensional extrapolation workflow on the device used here.

$(s_j, \lambda(s_j))$ on the one-dimensional joint extrapolation path, and the final estimate is obtained by a single extrapolation in $x = 1/N_{\text{Trotter}}$. Along this path, the learned-channel amplification supplies the physical-noise component of the paired node, while the inverse-step extrapolation removes the combined leading bias toward the joint zero-noise, zero-step target. The two parameter settings change both the target evolution angle and the depth-dependent balance between Trotter and noise effects, so agreement across the two panels supports the stability of the 1D joint error-mitigation workflow on the device used here.

b. Tensor-network validation. We also study the joint extrapolation path directly in a controlled tensor-network simulation using `quimb` [17]. **XZW: This is the clean validation of the coupled path, so give the full experimental contract before the conclusion: N, J, h, T, p, c, r-nodes, noise nodes, baseline grid size, reference value, and error metric.** The one-dimensional protocol evaluates the coupled curve $\lambda(s) = c(sT)^{p+1}$ with $p = 2$, **Xiaoyang: with $c = ?$ and $p = ?$,** while the baseline uses a tensor-product Richardson rule over independent step-size and noise-strength nodes. Thus the two-dimensional baseline provides a useful accuracy benchmark, but requires a rectangular grid of noisy circuit settings. As shown in Figure 4, the joint one-dimensional extrapolation has an evaluation error of 0.012, which is smaller than the two-dimensional baseline using a one-dimensional set of extrapolation nodes, and the measurement cost is reduced to one-fifth of the 2-d baseline. Therefore, the one-dimensional protocol preserves the extrapolation accuracy while reducing the measurement cost.

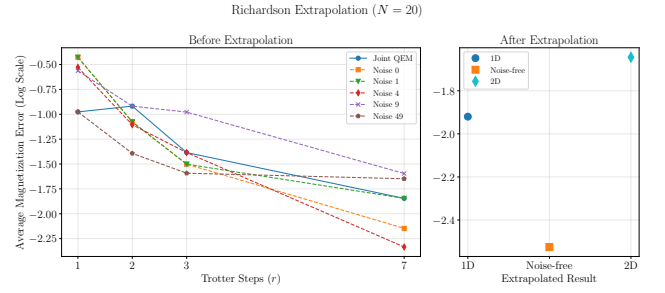


FIG. 4. Tensor-network validation on a 20-qubit one-dimensional Ising instance. Average magnetization error versus the number of Trotter steps. The one-dimensional joint method reduces the bias of noisy estimates and gives a final extrapolated value close to the two-dimensional Richardson baseline.

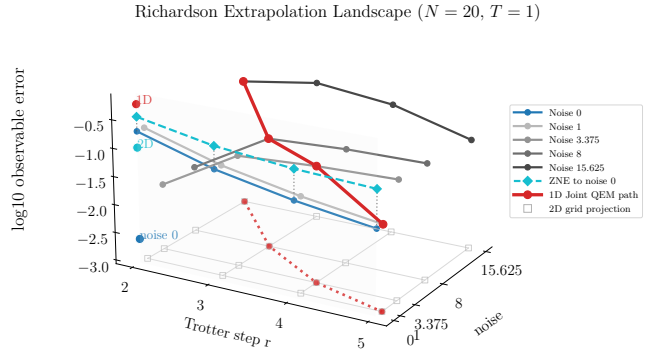


FIG. 5. Richardson extrapolation landscape for the simulated 20-qubit Ising instance. The observable error is plotted on a logarithmic scale over Trotter step counts $r = 2, 3, 4, 5$ and calibrated noise strengths for $J = h = T = 1$. Fixed-noise slices show the noisy data across Trotter step counts, cyan diamonds show zero-noise extrapolations at fixed r , and the red curve marks the one-dimensional joint-QEM path. Open squares indicate the projection of the two-dimensional Richardson grid used as the baseline.

II. DISCUSSION

The commutator analysis, hardware implementation, and numerical comparison indicate that physical noise can be incorporated into the same extrapolation variable that controls Trotter error when reliable tunable noise strengths are available. The remaining questions are practical: how to optimize the coupling constant and nodes for a given device, how accurately the effective Lindbladian can be learned at larger depths, and how the method behaves under non-Markovian or strongly correlated noise. These questions set the natural next scale-up path. The method is suitable for near-term and early fault-tolerant devices where increasing circuit depth is algorithmically urgent but physically unfeasible.

III. METHODS

XZW addressed: Methods should now serve as the audit trail for the claims in Results: node construction, cost comparison, hardware fitting choices, and tensor-network parameters. Keep theory details only insofar as they justify the extrapolation expansion.

a. Product-formula and noisy-channel model. Let $H = \sum_{\gamma=1}^{\Gamma} H_{\gamma}$ and suppose that a p -th order product formula implements one short-time step through

$$P(t) = \prod_{v=1}^M e^{-ita_v H_{\gamma_v}}. \quad (8)$$

Here H_{γ} are the Hamiltonian summands used by the product formula, γ_v specifies which summand appears in the v -th exponential and a_v is the corresponding product-formula coefficient, including the fractional weights and repetitions that occur in higher-order formulas. We model the noise after each Trotter layer as a Markovian Lindbladian process, so the implemented noisy layer is

$$\tilde{\mathcal{P}}(t, \lambda) = \prod_{v=1}^M (e^{\mathcal{L}_v} e^{-ita_v \text{ad}_{H_{\gamma_v}}}), \quad (9)$$

where λ is a tunable effective noise strength, for example supplied by probabilistic error amplification [1], and $\mathcal{L}_v$ is the layer noise generator. Operationally, λ denotes the calibrated noise strength supplied by the mitigation procedure: in the hardware workflow it is implemented by scaling an adaptively learned sparse Pauli-Lindblad channel rather than by increasing the Trotter circuit depth. For dimensionless step size s and total time T , the measured expectation is

$$g(s, \lambda) = \text{Tr} \left[O \tilde{\mathcal{P}}(sT, \lambda)^{1/s} \rho_0 \right]. \quad (10)$$

The target value is $\lim_{s, \lambda \rightarrow 0} g(s, \lambda)$.

b. Richardson nodes and extrapolation. For a scalar function

$$f(s) = f(0) + \sum_{\ell=1}^{m-1} a_{\ell} s^{\ell} + R_m(s), \quad (11)$$

an m -node Richardson rule estimates $f(0)$ by $\sum_{j=1}^m b_j f(s_j)$, where the coefficients satisfy $\sum_j b_j = 1$ and $\sum_j b_j s_j^{\ell} = 0$ for $1 \leq \ell \leq m-1$. Here s is the dimensionless inverse Trotter count; the physical duration of one product-formula step is sT . Executable nodes are therefore chosen by integer Trotter counts r_j through $s_j = 1/r_j$, so the circuit step duration is T/r_j . **XZW addressed:** Be careful with notation: earlier s behaves like a step size or inverse step number. If r_j is the Trotter count, define whether $s_j = T/r_j$ or $s_j = 1/r_j$ and use the same convention in every figure axis and fit. In the coupled protocol, the measured nodes are the pairs $(s_j, \lambda(s_j))$ with $\lambda(s) = c(sT)^{p+1}$. Equivalently,

the Richardson rule acts on the one-variable function $f(s) = g(s, \lambda(s))$ and extrapolates from the sampled nodes $s_j > 0$ to the ideal limit $s = 0$. The order m is the number of paired nodes and fixes how many low-order powers in this one-variable expansion are canceled. See Supplementary Note 1 [16] for the explicit node construction and coefficient bounds. The residual bias is bounded by $\|\mathbf{b}\|_1 \max_j |R_m(s_j)|$. The coupling constant c sets the relative scale of the amplified noise and can be chosen to balance the physical-noise and Trotter remainders. The exponent in $\lambda(s) = c(sT)^{p+1}$ is chosen so that the leading physical-noise contribution, which is linear in λ , enters the same one-variable expansion as the leading Trotter contribution. Choosing a smaller exponent would leave noise as the dominant error, whereas choosing a larger exponent would make the procedure behave like ordinary Trotter extrapolation with insufficient noise cancellation.

c. Finite-order BCH expansion. After imposing the coupled path, the one-step channel in Eq. (9) is the ordered BCH product $\tilde{\mathcal{P}}(sT, \lambda(s)) = \prod_{v=1}^M e^{\mathcal{K}_{2v}} e^{\mathcal{K}_{2v-1}}$, with generators

$$\mathcal{K}_{2v-1} = -isT a_v \text{ad}_{H_{\gamma_v}}, \quad \mathcal{K}_{2v} = c(sT)^{p+1} \mathcal{L}_v. \quad (12)$$

Instead of assuming that the full BCH series converges, we consider a finite-order truncation of the BCH expansion $e^{sT \mathcal{G}_{(q_0)}(s)}$ with the truncated effective generator:

$$sT \mathcal{G}_{(q_0)}(s) := -isT \text{ad}_H + \sum_{q=2}^{q_0} \mathcal{B}_q(\mathcal{K}_1, \dots, \mathcal{K}_{2M}), \quad (13)$$

$$\tilde{\mathcal{P}}(sT, \lambda(s)) = e^{sT \mathcal{G}_{(q_0)}(s)} + \mathcal{R}_{q_0+1}.$$

Here each $\mathcal{B}_q$ is the degree- q BCH Lie polynomial, a finite linear combination of q -fold nested commutators of the generators $\mathcal{K}_{\mu}$. The discarded tail $\mathcal{R}_{q_0+1}$ is then bounded directly using the truncation theorem of Ref. [15]. The tail is controlled by doubly right-nested commutator sums of the schematic form

$$\sum \left\| \left[[\mathcal{K}_{v_{1,1}}, \dots, \mathcal{K}_{v_{1,q_1}}], \dots, [\mathcal{K}_{v_{d,1}}, \dots, \mathcal{K}_{v_{d,q_d}}] \right] \right\|. \quad (14)$$

These sums upper-bound the truncation budget ε_{q_0} used in the Results: choosing q_0 and the step-size window so that the discarded BCH tail is at most ε_{q_0} gives $|f(s_j) - f_{q_0}(s_j)| \leq \varepsilon_{q_0}$ on the Richardson nodes. After substituting $\lambda(s) = c(sT)^{p+1}$, each BCH monomial is graded by its net power of s in the continuous-time effective generator. A Hamiltonian factor contributes one power of sT , a noise factor contributes $(p+1)$ powers through $\lambda(s)$, and passing from one step to total time divides by sT . Consequently, the ideal Hamiltonian term has grade zero, while the leading product-formula and physical-noise corrections both have grade p . Equivalently, before the total-time rescaling these corrections carry one additional one-step factor, which is why the channel-level coupling is written as $\lambda(s) \propto (sT)^{p+1}$ while the continuous-time effective generator is graded from

order p . **XZW addressed:** Add one reader bridge here: Results talks about leading observable bias of order s^{p+1} , while this generator paragraph says grade p after total-time rescaling. Explain the shift so the exponents do not look inconsistent. Richardson cancellation through grade $m-1$ therefore removes the two leading mechanisms with the same coefficients.

The remaining bias has two pieces: BCH monomials whose grade is not canceled by the chosen Richardson rule, and the finite-order truncation error from discarding terms above q_0 . For k -local Hamiltonians and local Lindbladian noise, nested commutators are nonzero only when successive local supports overlap. The commutator sums are therefore governed by support growth and local-extensiveness parameters, rather than by the extensive operator norm of the full Hamiltonian. Taking $q_0 = O(\log(N/\varepsilon))$ makes the truncation error $O(\varepsilon)$ while changing the allowed step size only by logarithmic local factors. For plane-wave electronic-structure Hamiltonians, the corresponding control parameter is the layer number of the second-quantized monomials generated under repeated commutation. The supplementary material gives the full truncation bookkeeping; here we use it only to justify the finite expansion on which the coupled Richardson rule acts.

d. Intrinsic noise floor. The theoretical path assumes that the experiment can realize the effective noise strengths $\lambda(s_j)$ at all Richardson nodes. Real hardware has an intrinsic noise level λ_0 before amplification, so the implemented values are better viewed as $\lambda_0 + \lambda_{\text{amp}}(s_j)$. If λ_0 is comparable to the smallest target value on the path, the uncanceled offset contributes a residual term that is not removed by the coupled Richardson coefficients. The resource statement therefore assumes that the floor is below the smallest coupled noise strength up to the target additive accuracy. In practice, one first chooses a set of Trotter counts and the accessible noise window $[\lambda_{\min}, \lambda_{\max}]$, then selects c so that

$$\lambda_{\min} \leq c(T/r_j)^{p+1} \leq \lambda_{\max} \quad \text{for all } j. \quad (15)$$

Equivalently, the admissible values of c are the intersection of the intervals $[\lambda_{\min}/(T/r_j)^{p+1}, \lambda_{\max}/(T/r_j)^{p+1}]$ over all nodes. If the intrinsic floor is written separately as λ_0 , the amplified component must realize $c(T/r_j)^{p+1} - \lambda_0$ within the available amplification range. **XZW addressed:** This should become an operational prescription: given a device noise window $[\lambda_{\min}, \lambda_{\max}]$ and chosen r_j , how is c selected or constrained? This also helps readers judge when the protocol is experimentally feasible. The same tunable noise strength is also required by a two-dimensional baseline; the coupled construction changes how many noise-strength values must be sampled.

e. Learned noise model. The hardware workflow uses an adaptively learned effective sparse Pauli-Lindblad model

$$\mathcal{L}(\rho) = \sum_{k \in \mathcal{K}} \lambda_k (P_k \rho P_k - \rho), \quad (16)$$

with $|\mathcal{K}|$ much smaller than the full Pauli group. The Pauli-Lindblad rates are obtained by adaptive learning on circuits matched to the target compiled circuit family. After learning, noise amplification is implemented by scaling the retained rates and sampling the corresponding Pauli insertions, which realizes calibrated noise strengths for ZNE without lengthening the target Trotter circuit. The learned channel is used as an effective model for the observable and circuit family; it is not intended to reconstruct every microscopic noise process. **XZW: Still needs hardware metadata: what circuits are used for learning, how many retained Pauli terms, and how the scaling factor maps learned rates to the amplified noise strengths used in Fig. 3.**

f. Hardware data processing. The measured observable is the average longitudinal magnetization, estimated from bitstrings produced by computational-basis measurements. Readout mitigation is applied before gate-noise learning by inverting local calibration matrices for the selected qubits at the observable level. For each paired node and calibrated noise-strength setting, repeated circuit executions provide sample means and uncertainty estimates. The node estimates are obtained with the same low-order polynomial or exponential calibration ansatz across all Trotter configurations within a panel, so that the one-dimensional path fit is not affected by changing the noise-amplification model from node to node. The paired values are then fitted as even polynomials in $x = 1/N_{\text{Trotter}}$ for the second-order symmetric formula. This hardware workflow implements the coupled one-dimensional path using learned-channel noise amplification; the tensor-network benchmark uses the same path in a controlled simulation where it can be compared directly with a two-dimensional Richardson grid. **XZW: Still needs hardware fit metadata: specify the exact fit models used in the plotted hardware panels, how the model/order is chosen, and how uncertainty is propagated from ZNE into the final $1/N_{\text{Trotter}}$ fit.**

g. Classical simulations. The tensor-network data use the same average magnetization observable as the hardware experiment, but on a 20-qubit open one-dimensional transverse-field Ising chain with $J = h = 1$ and total time $T = 1$. The initial state is the computational-basis state $|0\rangle^{\otimes N}$. Noisy and noiseless product-formula circuits are represented with quimb and sampled with a fixed random seed, using 10^4 computational-basis samples for each noisy circuit setting. The simulated Trotter counts are $r \in \{2, 3, 4, 5\}$. The noise grid is $\lambda \in \{0, 1, 3.375, 8, 15.625\}$, corresponding to the nonzero scale factors $(r/2)^3$ for $r = 2, 3, 4, 5$. The one-dimensional benchmark uses the paired monotone path

$$(r, \lambda) \in \{(2, 15.625), (3, 8), (4, 3.375), (5, 1)\},$$

while the two-dimensional baseline uses the full rectangular grid of four Trotter counts and five noise strengths. The plotted error is the logarithm of the absolute difference between the average magnetization and a com-

mon noiseless reference, computed from a sparse high-resolution evolution with 10^4 Taylor steps and then divided by N . XZW addressed: This paragraph is still too thin for a numerical benchmark. List J, h, T, initial state, p, c, r nodes, noise nodes, bond dimension or

truncation tolerance, and the noiseless reference used to compute the plotted error. Both methods are compared against this same reference expectation, so the plotted errors isolate the bias removed by extrapolation rather than differences in state preparation or measurement.

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